Beyond Accuracy: Trustworthy Index and Robustness Index for Evaluating CNN Reliability

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ABSTRACT

TBD

1. Introduction

We need to explain CNN reliability and XAI from title.

Despite excellent accuracy, a model may not be reliable. Explainability methods, such as Grad-CAM, show that a model may rely mainly on background and irrelevant regions instead of focused objects [**? ? ?**]. This raises questions about reliability and trustworthiness. we are not working to imrpove relaiability??

Traditional CNN model evaluation meausres fail to indicate the reliability of the correspoinding model.

To address this limitation (we did not say the limitation yet), we have proposed two novel evaluation metrics: Trustworthy Index (TI) and Robustness Index (RI) using explainable AI (XAI) methods. We highlight the importance of object reliability (is this correct? or model reliability?) as well as accuracy. TI integrates accuracy and object reliability (??) in clean images. RI measures the robustness under various perturbations. need to define robustness with ref Robustness measures how close the model's accuracy and explanations (object reliance) under perturbation remain to the outputs on clean images (good but not clear - remember if you have to think to understand a sentence - you should probably rewrite it!).

The proposed measures TI and RI rely on the fact that *how much a model's predictions rely on real objects instead of the background*. We introduce a new score named LMOS (Localization Mask Overlap Score) to indicate this fact.

As red and yellow regions of the Grad-CAM visualization have stronger influence than other regions for predictions, we consider them as decision-relevant. why do we need this sentence?

LMOS defines when the model relies on something to make decisions, what ratio of that reliance falls on focus objects. not clear

A model has high accuracy but low LMOS may be affected by shortcut learning, overfitting, bias, and reliance on background features. can we show this - i.e., a model having high acc but low LMOS and has one or more of the mentioned problems? **A model is trustworthy if it has high accuracy and high LMOS. Similarly, a model is considered robust under perturbations if it maintains high accuracy stability, and high LMOS stability under perturbations.**

need clear distinctions between EI and RI - what EI shows must be different than RI

Models with high TI is considered reliable as both the model accuracy and object reliance are high. Models with moderate to low TI means its accuracy or object focus, or both, are low and limited.

To evaluate the robustness of the models, we have applied nine perturbations (noise, blur, hue, occlusion, etc.) to the test images and then calculated accuracy and LMOS scores after each perturbation.

The mean LMOS stability under a perturbation has been obtained by averaging the image-wise LMOS stability values across all test samples. Then, ERI under a perturbation has been calculated as the mean of both stabilities. Finally, the mean ERI across all perturbations has been calculated and a ranking has been made according to the mean ERI. Like ETI, a model with hight ERI indicates that both its accuracy and object focus remain stable under the use of perturbations. A moderate/low ERI reveals that its accuracy, or focus on object, or both change with the use of perturbations.

Two questions that the proposed measures can answer - How trustworty a model is? Trustworty means - predictions are based on real objects in the images. How robust a model is? Robustness means - performance of the model will not degrade under purturbations.

To show in exp:

- Acc high but not relaiable - may be low robustness or predictions based on BG.
- a model having high acc but low LMOS and has one or more of the mentioned problems
- A model is trustworthy if it has high accuracy and high LMOS - show this. Idea: models with high ACC an high LMOS making more predictions based on objects in images rather than BGs

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- Show LMOS still works with purturbations and ACC and other traditional measure fails.
- Justify distinctions between EI and RI
- Show - traditional measures fail under purturbations but proposed measures hold.

2. Related Works

3. System Architecture

4. Materials and Methods

4.1. Dataset

4.2. Dataset Distribution

4.3. Training Configuration

4.4. Classification Performance

4.5. Creating JSON files of test images

4.6. Adding Perturbations

4.6.1. Gaussian Blur

4.6.2. Gaussian Noise

4.6.3. Brightness Adjustment

4.6.4. Hue Shift

4.6.5. Contrast Adjustment

4.6.6. Gamma Correction

4.6.7. Motion Blur

4.6.8. JPEG Compression

4.6.9. Random Occlusion

4.7. Grad-CAM Visualization

4.8. Necessary Scores Definition

4.8.1. LMOS (Localization Mask Overlap Score)

4.8.2. Accuracy Stability

4.8.3. LMOS Stability Under Perturbations

4.9. ETI, Explainable Trustworthy Index

4.9.1. Ablation Study

4.10. Explainable Robustness Index (ERI)

4.10.1. Ablation Study

5. Result and Discussion

5.1. Grad-CAM Visualization

5.1.1. Threshold Justification

5.2. Explainable Trustworthiness Index (ETI)

5.3. Explainable Robustness Index (ERI)

5.3.1. EgyPLI

5.3.2. BDMediLeaves

5.3.3. Flowers

5.3.4. Flowers Recognition KAGGLE Dataset

5.4. Cross-Dataset Analysis

6. Limitations

7. Future Work

8. Conclusion

Data and Code availability

Supplementary code and data (including `.json` files and `.keras` files) associated with this research are available at:
`https://github.com/ragib1996/ETI_ERI`

CRediT authorship contribution statement

Ragib Mehedi: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – Original Draft, Writing – Review & Editing. **Azmain Inquaid Haque:** Investigation, Data Curation, Software, Validation. **Md. Nazmul Abdal:** Data Curation. **Abdullah Al Noman:** Supervision, Writing – Review & Editing. **Md. Zahidul Islam:** Supervision, Writing – Review & Editing .

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References